PinchType: Text Entry for Virtual and Augmented Reality Using Comfortable Thumb to Fingertip Pinches

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Abstract

Text entry is an integral component to many use cases in virtual and augmented reality. We present PinchType: a new method of virtual text entry that combines users' existing knowledge of the QWERTY keyboard layout with simple thumb and finger interactions. Users pinch with the thumb and fingertip to select from the same group of letters the finger would press on a QWERTY keyboard; a language model disambiguates. In a preliminary study with 14 participants, we investigated PinchType's speed and accuracy on initial use, as well as its physical comfort relative to a mid-air keyboard. After entering 40 phrases, most people reported that PinchType was more comfortable than the mid-air keyboard. Most participants reached a mean speed of 12.54 WPM, or 20.07 WPM without the time spent correcting errors. This compares favorably to other thumb-to-finger virtual text entry methods.

Author Keywords

pinch typing; text entry; mixed reality

CCS Concepts

•**Human-centered computing** → **Text input;** *Mixed / augmented reality;*

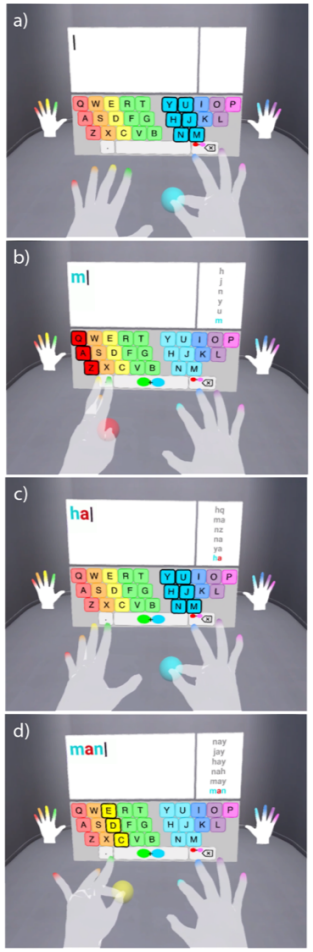


Figure 1: Typing "hand" with PinchType. Four successive pinches are made, corresponding to key groups that contain the letters in "hand". Notice how the first letter changes as each letter is added. Selection is in Fig. 2.

Introduction

As people begin adopting AR/VR for work and communication, text entry becomes increasingly important. Hand tracking is a valuable technology that can let users enter text without requiring a controller. Some methods of hand-based text entry [3, 9, 10, 14] involve making gestures in midair, including interacting with a floating virtual keyboard. These can achieve relatively high performance, but can also be fatiguing or cumbersome. Other techniques, which we will call "thumb-to-finger" techniques [2, 6, 7, 11, 12, 13], depend on one's thumb touching certain spots on the same hand. These can be more comfortable, can often be used eyes-free, and provide self haptic feedback of character entry events. But they can also be more complicated, unnatural, or difficult to learn. Meanwhile, language models are used to help people type quickly with fewer than 26 distinct alphabetic commands [1, 4, 11, 13] or to correct mistakes in decoding [14]. PinchType is a thumb-to-finger technique that leverages a language model to reduce the number of distinct gestures needed to cover the alphabet.

Design and Implementation

PinchType is based on the idea that experienced touch-typists have internalized the association between the letters of the alphabet and the fingers used to type them on a QWERTY keyboard (for example, the letters R, T, F, G, V, and B are associated with the left index finger while the letters I and K are associated with the right middle finger). To type a letter, the user simply pinches the finger associated with that letter to their thumb and releases. Given that multiple letters are associated with a given finger, we employ a dictionary to limit the set of possible words produced by a series of pinches. Our rudimentary bi-gram language model sorts this set of words by likelihood, and the user can choose a different word if their intended word was not the most likely option. Fig. 1 illustrates the PinchType flow.

We created our PinchType VR prototype using Unity 2018. To evaluate the effectiveness of the keyboard design independently from the limitations of any one particular hand tracking technique, we use a high-quality marker-based hand tracking system [5].

In addition to the basic interaction of pinching a single finger to select a group of letters, the following interactions were implemented:

- Backspace - Simultaneously pinch both pinky fingers
- Space - Simultaneously pinch both index fingers
- Word Disambiguation - Hold both index finger pinches and raise/lower the right hand

Inspired by mobile phone virtual keyboards, the first word of every sentence is automatically capitalized, typing two spaces in a row automatically adds a period to the end of the previous word, and holding the backspace gesture for half a second deletes an entire word. Numbers, symbols, and other action keys are not currently implemented, but there are many possible gestures that could map to them or change the "mode" of the keyboard.

As the user is typing a word, the language model's "most likely" word appears in the text area. Each of its letters are colored according to their corresponding letter group. There is also a sidebar next to the text area that displays the language model's top five predicted words. If, when the user is done typing a word, the model correctly chose it as most likely, the user can simply make the gesture for *space* (i.e., pinch with both index fingers) to "lock in" this word. If the word they intended was not chosen but is listed in the sidebar, they can hold the space gesture, raise their right hand until the intended word is highlighted, and then release the pinches to "lock in" that word. Fig. 2 illustrates the disambiguation process.



Figure 2: Selecting the word "hand" with PinchType. After the correct series of pinches, the language model chose "jane" (e), and the user disambiguates (f-g).

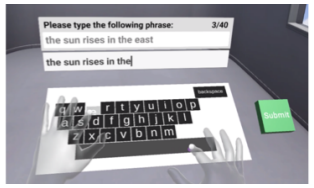


Figure 3: A participant uses PokeType to transcribe a phrase.

User Study

We conducted a preliminary user study focusing on PinchType's effectiveness, comfort, and ease on initial use. We recruited 14 participants who all self reported as touch typists. We showed each participant the PinchType prototype, explained to them how it worked, and asked them to type five practice phrases. Participants then typed 40 phrases as quickly and accurately as possible.

We used a subset of phrases from MacKenzie and Soukoreff's Phrase Sets for Evaluating Text Entry Techniques [8]. Our subset included only phrases that had under 27 characters and no words longer than seven characters, to keep both the phrase and individual word difficulty consistent.

We also had each participant try a more familiar method of virtual text entry that we call PokeType – a midair QWERTY keyboard (Fig. 3). Pressing on a letter key with either index finger types that letter, like traditional hunt-and-peck typing. Dudley et al. [3] identified two-finger midair typing like this to be particularly effective, and we consider PokeType a reasonable upper bound on performance. Roughly half of the participants tried PinchType first while the rest tried PokeType first, following the same procedure. After each method of typing, they were given a questionnaire about the amount of mental and physical energy required to type, and the comfort of their hands and arms.

Results and Discussion

Typing Performance

For each keyboard, each participant's 40 phrases were split into 4 separate blocks of 10 to track their progress over time. Their typing speed in Words Per Minute (WPM), Corrected Error Rate (CER), and Uncorrected Error Rate (UER) were calculated for each block, and the mean results are shown below (Fig. 4). We analyze the performance data from 11 participants, as 3 did not complete all blocks for both typing methods in the allotted time. Our calculations are shown in Fig. 7-10.

The mean PinchType speed in the first block was 9.16 WPM and increased to 12.54 WPM in the fourth block. The mean PokeType speed in the first block was 21.51 WPM and increased to 25.42 WPM in the fourth block. A two-way RM-ANOVA found significant main effects for typing method ($p<0.001$) and for blocks ($p<0.001$), but no interaction effect ($p=0.065$). It is not surprising that participants could PokeType more quickly than they could PinchType; this midair two-finger method is one of the fastest virtual typing methods we could find in the literature [3]. A mean speed of 12.54 WPM for PinchType after 40 phrases (an average of 26 minutes) of practice is a promising result for a thumb-to-finger typing method.

For comparison, KITTY [7] had a mean speed of 20 strokes per minute over one session, which converts to 4 WPM. DigiTouch [12] participants had a mean typing speed of 7.0 WPM after one 20-minute session and 16.0 WPM after ten 20-minute sessions. HiFinger [6] had a mean speed of 9.82 WPM after six forty-word sessions. Both KITTY and HiFinger allow access to numbers and special keys that PinchType currently doesn't support, and this added complexity could contribute to their slower speeds. However, we believe that adding PinchType support for such keys wouldn't involve adding any complexity to the existing letter-typing process. As far as we could find, PinchType's 12.54 WPM compares favorably to other thumb-to-finger methods with a similar amount of practice.

The mean CER decreased over time for PinchType from 19.7% (2.3 times the mean CER for PokeType) to 13.3% (1.5 times that of PokeType). The mean UER was below 2% for both keyboards. Two-way ANOVAs found a signifi-

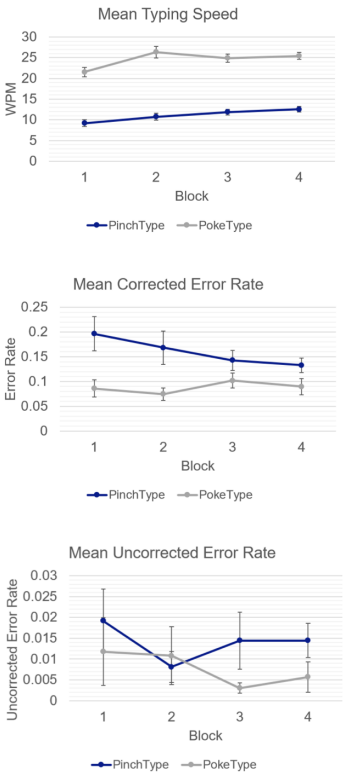


Figure 4: Line graphs depicting the mean typing speed (top), corrected error rate (middle), and uncorrected error rate (bottom) for both PinchType and PokeType over the four time blocks (with standard error bars).

cant main effect between typing methods for CER ($p<0.001$), but not for UER ($p=0.539$). There were no significant main effects for blocks and no interaction effects for either error metric. Though block number did not have a statistically significant effect on PinchType CER, we note that two participants started with very high CERs in the first PinchType block (36.5% and 42.2%) and were able to lower their CERs (to 13.3% and 6.9%) by block four.

Error identification and correction when using PinchType can be very time consuming. When the user makes an incorrect pinch or misses a pinch, it's not immediately clear that this has happened, because even making a correct pinch does not always result in the correct letter until the word is complete. Only when the participant reaches the end of the word will it be clear that there has been a mistake. From here, it can be hard to see the location of the error. We observed that this was a major obstacle for participants. When they made a mistake, they often didn't notice until later, and either backspaced very slowly through the word or deleted the entire word and started over. This can have a major impact on one's words per minute.

For this reason, we have also calculated a "perfect word" typing speed for PinchType, which roughly represents typing speed without the time spent correcting errors. We define a "perfect word" to be a word typed without ever using backspace. In other words, this value is the same as our normal speed calculation except we only count the time and characters from perfect words. The mean perfect word speed over each block is plotted above (Fig. 5), along with the standard WPM measurement. On average, the perfect word speed is 1.57 times faster than the raw typing speed, and it increases steadily over the course of the study, ending at 20.07 WPM. Though it is a little unrealistic to consider a situation where error correction is never needed, this

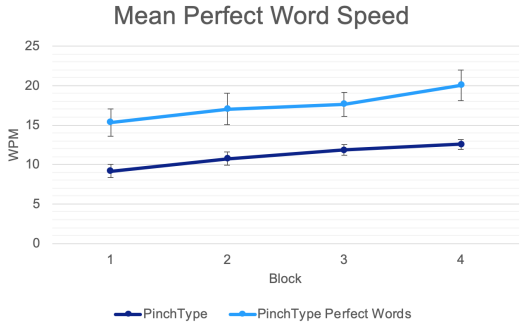


Figure 5: Line graph depicting the mean typing speed and mean perfect word typing speed for PinchType over the four time blocks.

metric at least suggests that reducing the amount of time needed for error correction–perhaps by improving the language model or UI–would significantly improve PinchType performance.

Qualitative Feedback

As part of the questionnaire for each text entry method, participants were asked "How comfortable do your hands and arms feel after using this keyboard?" They chose from five options ranging from "Very Uncomfortable" to "Very Comfortable." The results are shown in Fig. 6. Eight of the participants rated PinchType as more comfortable than PokeType, three rated PokeType as more comfortable than PinchType, and three rated them the same. None of the participants rated PinchType as being uncomfortable at all, whereas seven rated PokeType as uncomfortable. Regarding physical and mental energy, seven participants rated PokeType as requiring more physical energy, four said PinchType required more, and three rated them the same. Thirteen participants reported that PinchType required more mental energy, while one rated them the same. Participants'

$$WPM = \frac{SSL - 1}{Time} \times \frac{60}{5}$$

Figure 7: WPM Calculation
SSL: Submitted String Length

$$CER = \frac{CharDeletions}{TargetCharCount}$$

Figure 8: CER Calculation

$$UER = \frac{LD}{max(SL, TL)}$$

Figure 9: UER Calculation
LD: Levenshtein Edit Distance of Submitted and Target Text
SL: Submitted Text Length
TL: Target Text Length

$$PW = \frac{TPL - NPW}{PWTime} \times \frac{60}{5}$$

Figure 10: Perfect WPM Calculation
TPL: Total Length of Perfect Words
NPW: Number of Perfect Words
PWTime: Total Time Spent Typing Perfect Words

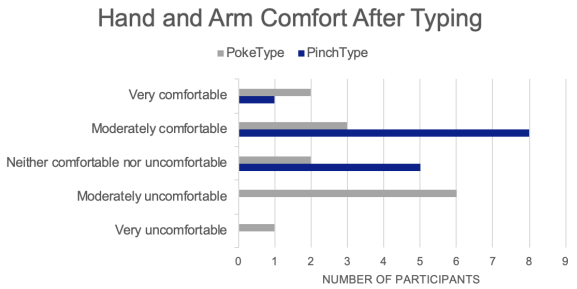


Figure 6: Bar graph depicting the number of participants that gave each comfort rating (on a five-point scale from "Very Uncomfortable" to "Very Comfortable") for both typing methods.

short-answer comments were consistent with the data: they mostly found PinchType less physically taxing but PokeType more intuitive.

Future Work

Our initial study's results were promising, supporting the idea that PinchType is a comfortable method of text input, and suggesting that its performance could be competitive with other similar input methods. However, our study lasted only a single session. Typing speed and accuracy did not plateau for most participants within their first 40 phrases, suggesting that there is still opportunity for better performance with more practice. A study over many sessions would better reveal peak PinchType performance.

A longer study could also determine how much experience is needed to use PinchType eyes-free. The authors of this paper (who have significantly more than 40 phrases worth of practice) can use PinchType without the keyboard UI, which indicates that this is possible, but an interface specifically for an eyes-free use case has yet to be developed.

Pursuing this as a next step could demonstrate many potential advantages of PinchType: users could type with their hands lowered (possibly increasing comfort even more), when walking, or when visual real estate is limited.

Our study also revealed several changes that could improve performance. The first is the robustness of the language model and disambiguation tool. Though the model usually chooses the intended word when the user makes the correct series of pinches, it has no margin for error. Suggestions with autocorrection in the word list could improve performance. Additionally, it's worth exploring different UI designs that might make error correction more intuitive and efficient. The current colored letters don't seem to be sufficient for users to identify the location of their mistake.

Conclusion

PinchType is a new method of bare hand text entry for virtual reality. A user can select groups of letters from the QWERTY keyboard by touching the appropriate fingertip to their thumb, and a language model helps them disambiguate their intended word. We conducted a preliminary study to evaluate this typing method, using a two-finger midair poke keyboard as a point of comparison. Qualitative feedback suggests that PinchType is comfortable, like other thumb-to-finger-based methods, and mean typing speeds indicate that it may be more effective than similar methods. Participants were able to reach an average speed of 12.54 WPM in a single session, and we expect that improvements to reduce the amount of time spent correcting errors would significantly improve this. There is likely no single "best" method of text entry in augmented or virtual reality, but for some use cases, PinchType's combination of comfort and effectiveness has the potential to improve people's work, play, and communication.

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